

Discrete
Mathematics
Segment One



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1. Introduction to Set

A *set* is an unordered collection of objects, called *elements* or *members* of the set. A set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

EXAMPLE 1 The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.

EXAMPLE 2 The set O of odd positive integers less than 10 can be expressed by $O = \{1, 3, 5, 7, 9\}$.

1. Introduction to Set

Two sets are *equal* if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter. Note also that it does not matter if an element of a set is listed more than once, so $\{1, 3, 3, 3, 5, 5, 5, 5\}$ is the same as the set $\{1, 3, 5\}$ because they have the same elements. 

1. Introduction to Set

There is a special set that has no elements. This set is called the **empty set**, or null set, and is denoted by $\emptyset$. The empty set can also be denoted by $\{\}$.

A set with one element is called a **singleton set**. A common error is to confuse the empty set $\emptyset$ with the set $\{\emptyset\}$, which is a singleton set.

1. Introduction to Set

Sets can be represented graphically using **Venn diagrams**, named after the English mathematician John Venn, who introduced their use in 1881.

In Venn diagrams the **universal set U** , which contains all the objects under consideration, is represented by a rectangle.

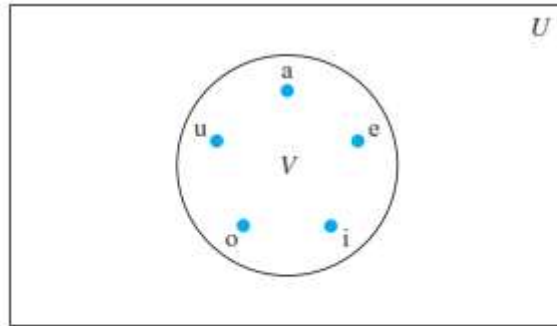


FIGURE 1 Venn Diagram for the Set of Vowels.

1. Introduction to Set

The set A is a *subset* of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

Example:

- The set of all odd positive integers less than 10 is a subset of the set of all positive integers less than 10.
- The set of all computer science majors at your school is a subset of the set of all students at your school.

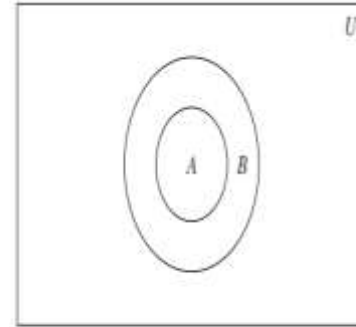


FIGURE 2 Venn Diagram Showing that A Is a Subset of B .

1. Introduction to Set

For every set S , (i) $\emptyset \subseteq S$ and (ii) $S \subseteq S$.

Proof: We will prove (i) and leave the proof of (ii) as an exercise.

Let S be a set. To show that $\emptyset \subseteq S$, we must show that $\forall x(x \in \emptyset \rightarrow x \in S)$ is true. Because the empty set contains no elements, it follows that $x \in \emptyset$ is always false. It follows that the conditional statement $x \in \emptyset \rightarrow x \in S$ is always true, because its hypothesis is always false and a conditional statement with a false hypothesis is true. Therefore, $\forall x(x \in \emptyset \rightarrow x \in S)$ is true. This completes the proof of (i). Note that this is an example of a vacuous proof. 

1. Introduction to Set

When we wish to emphasize that a set A is a subset of a set B but that $A \neq B$, we write $A \subset B$ and say that A is a **proper subset** of B .

A set is said to be **infinite** if it is not finite.

Example: The set of positive integers is infinite.

1. Introduction to Set

The cardinality of Set: [Aut'23]

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S . The cardinality of S is denoted by $|S|$.

Remark: The term *cardinality* comes from the common usage of the term *cardinal number* as the size of a finite set.

Let A be the set of odd positive integers less than 10. Then $|A| = 5$. 

Let S be the set of letters in the English alphabet. Then $|S| = 26$. 

Because the null set has no elements, it follows that $|\emptyset| = 0$. 

1. Introduction to Set

Given a set S , the power set of S is the set of all subsets of the set S . The **power set** of S is denoted by $P(S)$.

Example: What is the power set of the set $\{0, 1, 2\}$?

Solution:

The power set $P(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$.
Hence, $P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$.

1. Introduction to Set

Let A and B be sets. The *Cartesian product* of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

1. Introduction to Set

What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?

Solution: The Cartesian product $A \times B \times C$ consists of all ordered triples (a, b, c) , where $a \in A$, $b \in B$, and $c \in C$. Hence,

$$A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), \\ (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}.$$

1. Introduction to Set

Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.

Solution:

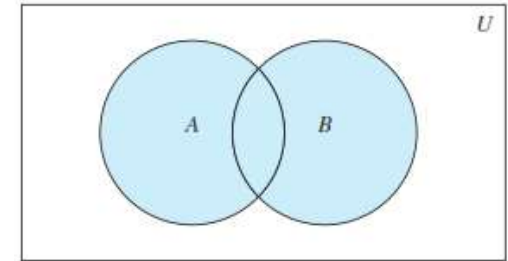
The only difference between $A \times B \times C$ and $(A \times B) \times C$ is parentheses, so for all practical purposes one can think of them as essentially the same thing. By Definition 9, the elements of $A \times B \times C$ consist of 3-tuples (a, b, c) , where $a \in A$, $b \in B$, and $c \in C$. By Definition 8, the elements of $(A \times B) \times C$ consist of ordered pairs (p, c) , where $p \in A \times B$ and $c \in C$, so the typical element of $(A \times B) \times C$ looks like $((a, b), c)$. A 3-tuple is a different creature from a 2-tuple, even if the 3-tuple and the 2-tuple in this case convey exactly the same information. To be more precise, there is a natural one-to-one correspondence between $A \times B \times C$ and $(A \times B) \times C$ given by $(a, b, c) \leftrightarrow ((a, b), c)$.

2. Set operation

Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

EXAMPLE:

- The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$
- The union of the set of all computer science majors at your school and the set of all mathematics majors at your school is the set of students at your school who are majoring either in mathematics or in computer science (or in both).



$A \cup B$ is shaded.

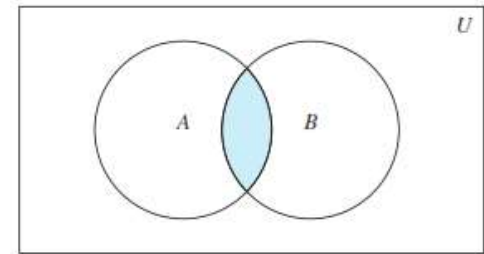
FIGURE 1 Venn Diagram of the Union of A and B .

2. Set operation

Let A and B be sets. The *intersection* of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

EXAMPLE:

- The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 3\}$; that is, $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$.
- The intersection of the set of all computer science majors at your school and the set of all mathematics majors is the set of all students who are joint majors in mathematics and computer science.



$A \cap B$ is shaded.

FIGURE 2 Venn Diagram of the Intersection of A and B .

2. Set operation

Two sets are called *disjoint* if their intersection is the empty set.

Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Because $A \cap B = \emptyset$, A and B are disjoint. 

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The generalization of this result to unions of an arbitrary number of sets is called the **principle of inclusion-exclusion**. The principle of inclusion-exclusion is an important technique used in enumeration.

2. Set operation

Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the *complement of B with respect to A* .

EXAMPLE 6 The difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{5\}$; that is, $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$. This is different from the difference of $\{1, 2, 3\}$ and $\{1, 3, 5\}$, which is the set $\{2\}$. ◀

EXAMPLE 7 The difference of the set of computer science majors at your school and the set of mathematics majors at your school is the set of all computer science majors at your school who are not also mathematics majors. ▶

2. Set operation

Let U be the universal set. The *complement* of the set A , denoted by $\overline{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.

EXAMPLE 8 Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then $\overline{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$. 

EXAMPLE 9 Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Then $\overline{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. 

2. Set operation

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

Solution: The set $A \cup B \cup C$ contains those elements in at least one of A , B , and C .

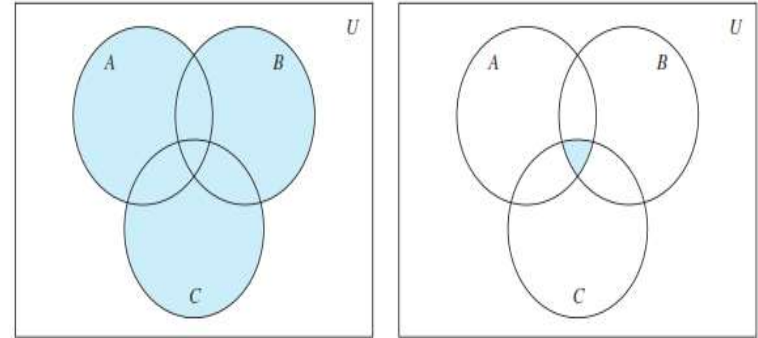
Hence,

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}.$$

The set $A \cap B \cap C$ contains those elements in all three of A , B , and C .

Thus,

$$A \cap B \cap C = \{0\}.$$



(a) $A \cup B \cup C$ is shaded.

(b) $A \cap B \cap C$ is shaded.

FIGURE 5 The Union and Intersection of A , B , and C .

2. Set operation

Example: In a survey, there are 90 students, it was found that 50 had taken C++, 35 had taken Discrete mathematics and 45 had taken Competitive programming. 15 had taken C++ and Discrete mathematics, 20 had taken Discrete mathematics and Competitive programming, 18 had taken C++ and Competitive programming and 10 had taken all three subjects. [Aut'23]

Find the number of students who had

- i) read exactly two subjects
- ii) read none of the subjects
- iii) read exactly one subject

2. Set operation

Example: In a survey, there are 90 students, it was found that 50 had taken C++, 35 had taken Discrete mathematics and 45 had taken Competitive programming. 15 had taken C++ and Discrete mathematics, 20 had taken Discrete mathematics and Competitive programming, 18 had taken C++ and Competitive programming and 10 had taken all three subjects. [Aut'23]

Find the number of students who had

- i) read exactly two subjects
- ii) read none of the subjects
- iii) read exactly one subject

Solution:

Given,

Total number of students surveyed (universal set), $n(S) = 90$

Number of students who took C++, $n(C++) = 50$

Number of students who took Discrete mathematics, $n(DM) = 35$

Number of students who took Competitive programming, $n(CP) = 45$

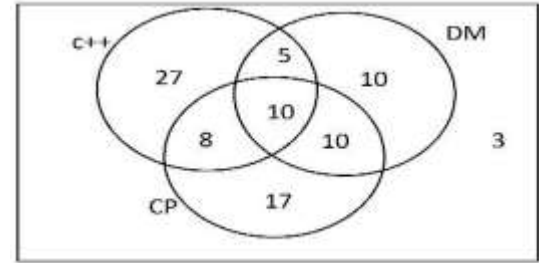
Number of students who took C++ and Discrete mathematics, $n(C++ \cap DM) = 15$

Number of students who took Discrete mathematics and Competitive programming, $n(DM \cap CP) = 20$

Number of students who took C++ and Competitive programming, $n(C++ \cap CP) = 18$

Number of students who took all three subjects, $n(C++ \cap DM \cap CP) = 10$

We can use the principle of inclusion-exclusion to find the number of students who fall into each category:

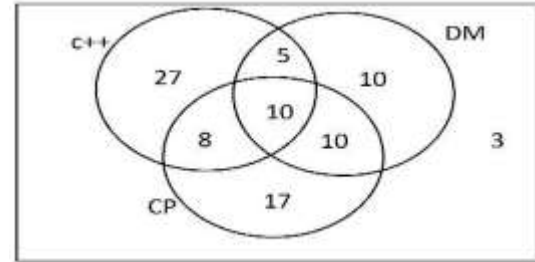


2. Set operation

i) Number of students who read exactly two subjects:

$$\begin{aligned}\text{Only DM and C++} &= n(\text{C++} \cap \text{DM}) - n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 15 - 10 \\ &= 5 \\ \text{Only DM and CP} &= n(\text{DM} \cap \text{CP}) - n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 20 - 10 \\ &= 10 \\ \text{Only C++ and CP} &= n(\text{C++} \cap \text{CP}) - n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 18 - 10 \\ &= 8\end{aligned}$$

Total number of students read exactly two subjects = $5 + 10 + 8 = 23$



2. Set operation

ii) Number of students who read none of the subjects:

Students who took at least one subject, $n(C++ \cup DM \cup CP)$

$$= n(C++) + n(DM) + n(CP) - n(C++ \cap DM) - n(DM \cap CP) - n(C++ \cap CP) + n(C++ \cap DM \cap CP)$$

$$= 50 + 35 + 45 - 15 - 20 - 18 + 10$$

$$= 87$$

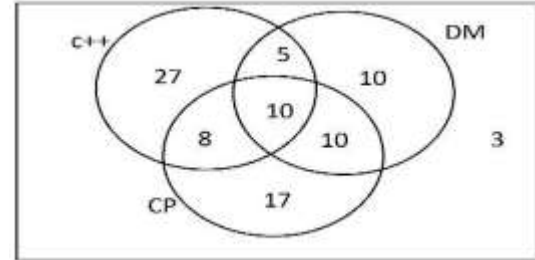
Students who read none of the subjects

= Total students - Students who took at least one subject

$$= n(S) - n(C++ \cup DM \cup CP)$$

$$= 90 - 87$$

$$= 3$$



2. Set operation

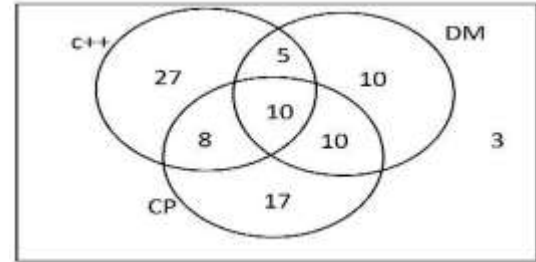
iii) Number of students who read exactly one subject:

$$\begin{aligned}\text{Only C++} &= n(\text{C++}) - n(\text{C++} \cap \text{DM}) - n(\text{C++} \cap \text{CP}) + n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 50 - 15 - 18 + 10 \\ &= 27\end{aligned}$$

$$\begin{aligned}\text{Only DM} &= n(\text{DM}) - n(\text{C++} \cap \text{DM}) - n(\text{DM} \cap \text{CP}) + n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 35 - 15 - 20 + 10 \\ &= 10\end{aligned}$$

$$\begin{aligned}\text{Only CP} &= n(\text{CP}) - n(\text{C++} \cap \text{CP}) - n(\text{DM} \cap \text{CP}) + n(\text{C++} \cap \text{DM} \cap \text{CP}) \\ &= 45 - 18 - 20 + 10 \\ &= 17\end{aligned}$$

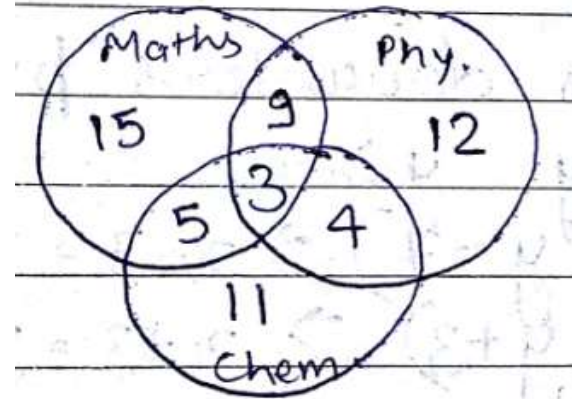
Students who took exactly one subject = $27 + 10 + 17 = 54$



2. Set operation

Example: In a survey of 25 students, it was found that 15 had Mathematics, 12 had taken physics and 11 had taken Chemistry, 5 had taken Mathematics and Chemistry, 9 had taken Mathematics and Physics, 4 had taken Physics and Chemistry and 3 had taken all the three subjects. Find the number of students that had

- (i) Only Chemistry
- (ii) Physics and Chemistry but not mathematics
- (iii) Only one of the subjects
- (iv) None of the subjects
- (v) at least one of the three subjects



2. Set operation

(i) Only Chemistry : $11-5-4+3=5$

(ii) Physics and Chemistry but not mathematics : $4-3=1$

(iii) Only one of the subjects :

Only Mathematics: $15-5-9+3 = 4$

Only Physics: $12-9-4+3 = 2$

Only Chemistry: $11-5-4+3=5$

Total: 11

(iv) None of the subjects:

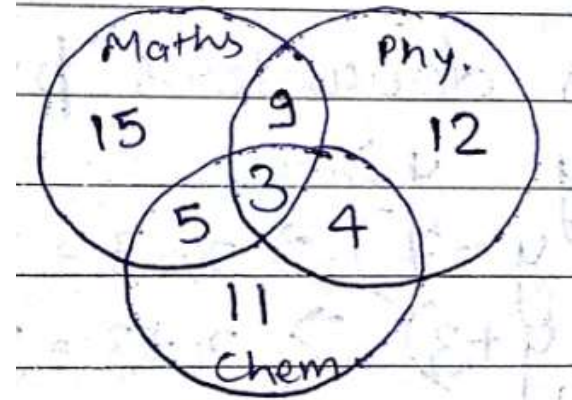
At least One: $11+9+5+4-2(3)$

$$= 29-6$$

$$= 23$$

None: $25-23 = 2$

(v) at least one of the three subjects: 23



3. Representation of Sets

Sets can be represented in two ways –

- Roster or Tabular Form
- Set Builder Notation

3. Representation of Sets

Roster or Tabular Form:

The set is represented by listing all the elements comprising it. The elements are enclosed within braces and separated by commas.

Example 1 – Set of vowels in English alphabet, $A=\{a,e,i,o,u\}$

Example 2 – Set of odd numbers less than 10, $B=\{1,3,5,7,9\}$

3. Representation of Sets

Set Builder Notation: [Aut'23]

The set is defined by specifying a property that elements of the set have in common. The set is described as $A = \{x:p(x)\}$

Example 1 – The set $\{a,e,i,o,u\}$ is written as –

$A = \{x:x \text{ is a vowel in English alphabet}\}$

Example 2 – The set $\{1,3,5,7,9\}$ is written as –

$B = \{x:1 \leq x < 10 \text{ and } (x\%2) \neq 0\}$

4. Algebraic Properties of Set

<i>Identity</i>	<i>Name</i>
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

4. Algebraic Properties of Set

Prove Associative Laws: $(A \cup B) \cup C = A \cup (B \cup C)$

Solution:

Let some $x \in (A \cup B) \cup C$

$\Rightarrow (x \in A \text{ or } x \in B) \text{ or } x \in C$

$\Rightarrow x \in A \text{ or } x \in B \text{ or } x \in C$

$\Rightarrow x \in A \text{ or } (x \in B \text{ or } x \in C)$

$\Rightarrow x \in A \text{ or } x \in B \cup C$

$\Rightarrow x \in A \cup (B \cup C).$

Similarly, if some $x \in A \cup (B \cup C)$, then $x \in (A \cup B) \cup C$.

Thus, any $x \in A \cup (B \cup C) \Leftrightarrow x \in (A \cup B) \cup C$. Hence Proved.

4. Algebraic Properties of Set

Prove De Morgan's Laws: $(A \cup B)' = A' \cap B'$

Solution:

To Prove $(A \cup B)' = A' \cap B'$

Let $x \in (A \cup B)' \Rightarrow x \notin (A \cup B)$

$(\because a \in A \Leftrightarrow a \notin A')$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \in A' \cap B'$

Therefore, $(A \cup B)' \subset A' \cap B'$ (i)

Again, let $x \in A' \cap B' \Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \notin (A \cup B)$

$\Rightarrow x \in (A \cup B)'$

Therefore, $A' \cap B' \subset (A \cup B)'$ (ii)

Combining (i) and (ii), we get $A' \cap B' = (A \cup B)'$. Hence Proved.

5. Computer representation of set

Method for storing elements using an arbitrary ordering of the elements of the universal set. Specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$. Represent a subset A of U with the bit string of length n , where the i^{th} bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A .

5. Computer representation of set

Example: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

i) What bit strings represent the subset of all odd integers in U ?

The bit string that represents the set of odd integers in U , $\{1, 3, 5, 7, 9\}$, has a one bit in the first, third, fifth, seventh, and ninth positions. It is 10 1010 1010.

ii) What bit strings represent the subset of all even integers in U ?

The bit string that represent the subset of even integers in U , $\{2, 4, 6, 8, 10\}$.

It is 01 0101 0101.

iii) What bit strings represent the subset of integers not exceeding 5 in U ?

The set of all integers in U that do not exceed 5, $\{1, 2, 3, 4, 5\}$, is represented by the string 11 1110 0000.

5. Computer representation of set

Example: The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 11 1110 0000 and 10 1010 1010.

Solution: The bit string for the **union** of these sets is
 $11\ 1110\ 0000 \vee 10\ 1010\ 1010 = 11\ 1110\ 1010$,
which corresponds to the set $\{1, 2, 3, 4, 5, 7, 9\}$.

The bit string for the intersection of these sets is
 $11\ 1110\ 0000 \wedge 10\ 1010\ 1010 = 10\ 1010\ 0000$,
which corresponds to the set $\{1, 3, 5\}$.

5. Computer representation of set

Example: Find the bitwise OR, bitwise AND, and bitwise XOR of the bit string 01 1011 0110 and 11 0001 1101.

Solution:

01 1011 0110

11 0001 1101

11 1011 1111

bitwise *OR*

01 0001 0100

bitwise *AND*

10 1010 1011

bitwise *XOR*

Table for the Bit Operators <i>OR</i> , <i>AND</i> , and <i>XOR</i> .				
x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0